Sardar Patel Institute of Technology

Munshi nagar, Andheri(west), Mumbai-400058

Mini Project : T.E. (EXTC) Sem: V Academic Year 2026-27

"Microcontroller-Based Cargo Vehicle Overload Protection System Using Discrete Switching Circuitry"

AMAR WARE | ARYAN KUPEKAR | SOHAM KEDARI | Prof. MILIND PARAYE (Mentor)

Introduction or Abstract

The Smart Cargo System is an embedded system designed to measure cargo weight, display real-time readings, and support overload protection. It combines four 50 kg square load cells connected in half-bridge manner, one HX711 amplifier module, an STM32 microcontroller, and an OLED display. Wi-Fi communication and overload-control hardware are being developed for remote monitoring and protection.

Project Goals

1.Measure cargo weight using load cells.

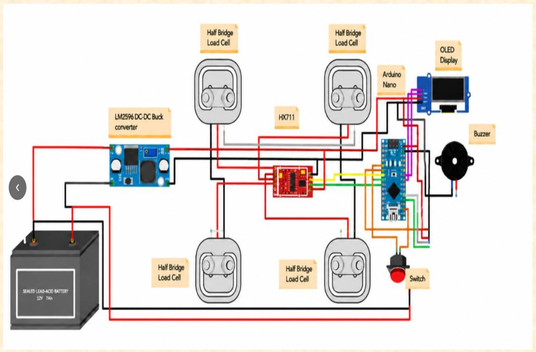
2.Display weight reading on OLED Display.

3.Detect Overload conditions exceeding the configured 60kg threshold.

4. Develop a discrete transistor-based relay circuit when overloaded is detected.

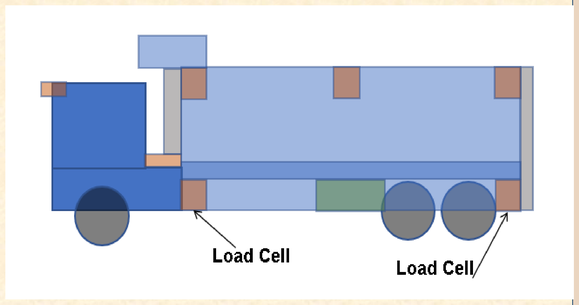
5. Enable wireless weight monitoring via a custom web dashboard.

Circuit Diagram

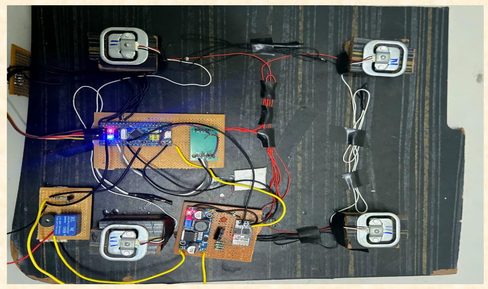


Project Description

This system is an embedded solution designed for real-time cargo weight measurement, local display, and overload protection using an STM32 microcontroller and an HX711 amplifier with load cells. It features an OLED display for immediate readings, integrated Wi-Fi telemetry via ESP8266 for web-based fleet monitoring, and automated relay-based protection to prevent vehicle overload.



Hardware Implementation





Findings

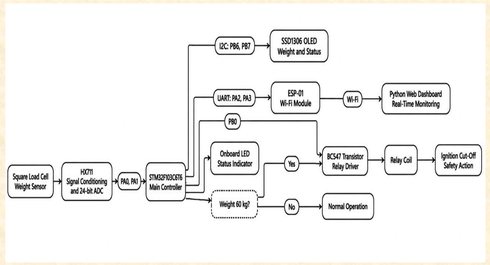
1.Freight vehicle overloading is a primary cause of severe road accidents, premature vehicle failure, and dangerous brake system degradation.

2.To address this critical issue, this project provides an automated embedded monitoring system that continuously tracks payload weight and directly interrupts system power via a safety relay whenever cargo exceeds the safe 60 kg limit.

3.By integrating Wi-Fi telemetry, the system streams live data to a web dashboard, giving logistics managers remote oversight to prevent illegal or dangerous loads before a vehicle hits the road.

Charts & Graphs





Conclusion

This project successfully demonstrates an embedded cargo-vehicle monitoring system designed to mitigate accident risks caused by vehicle overloading. By combining an STM32 microcontroller, HX711 amplifier, and OLED display, the system provides accurate real-time payload tracking. The addition of a discrete transistor relay driver ensures immediate hardware protection above 60 kg, while planned Wi-Fi telemetry via ESP8266 offers a scalable solution for remote logistics supervision and road safety compliance.

References

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[2] S. Shi, M. Yarnold, S. Hurlebaus, and J. Mander, "Evaluation of a Load Cell-Based Bridge Weigh-in-Motion System," Transportation Research Record: Journal of the Transportation Research Board, vol. 2679, no. 2, pp. 535–549, 2025. doi: 10.1177/03611981241263336.

[3] B. Jacob and L.-M. Cottineau, "Weigh-in-Motion for Direct Enforcement of Overloaded Commercial Vehicles," Transportation Research Procedia, vol. 14, pp. 1413–1422, 2016. doi: 10.1016/j.trpro.2016.05.214.

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